characteristic of thermo dynamic change, Rudolf Clausius (1822-88) introduced the concept of entropy.

In a broad sense, this expressed the degree to which the energy in a thermodynamic system was unavailable for use. Gibbs recognized the full significance of Clausius theory and used it to treat the way the physics states of homogeneous substances volume, energy and entropy varied with thermodynamic conditions. He employed geometrical methods of representing by diagram the thermodynamic states. Gibbs noted that his approach would work also for chemically mixed materials. That insight provided the seed for his masterpiece. On the Equilibrium, of Heterogeneous Substances (1878). This contained a simple mathematical formula which predicted the number of different physical combinations in which the component of the mixture could co-exist

SPEED PRACTICE – 5:

SPEED PRACTICE - 5 Time - 15 Minutes

Type the following in double line spacing with a margin of 10 degrees:-

As a wake up cup in the morning, as a pep up drink during the day and as an item of hospitality, coffee has worked its way into the marrow of civil8ised life. From a humble beginning as a food and drink for the African and Arabian tribesmen, coffee has evolved into a global phenomenon of unimaginable proportions. The colourful and interesting stories behind its discovery, propagation and use and its importance today are amazing. The legends and lores are spread across centuries and across the world. Coffee Arabica, popularly known as the Coffee plant, grew wild in Ethiopia, in Africa, millions of years ago.

Around the 6th century traders and pilgrims discovered that a grinding and mixing of coffee beans with animal fat made a sustainable snack during their arduous journey across the deserts from Ethiopia to Saudi Arabia. This concoction was made in the size of billiard balls and was carried in their saddle bags. These trave4llers were responsible for popularizing coffee in East Africa and Arabia. A few centuries later, a priest in an |Arabian monastery observed that those goats which consumed a lot of red coffee beans were more agile and friskier than others who did not. The priest thought that he could keep himself from falling asleep. During his evening prayers, if he could try consuming these berries. He conducted the first bewing experiment in his monastery by parching the beans, steeping them in water and drinking the liquid. The result was immediate and rewarding.

The priest and other monks religiously took this concoction with great enthusiasm. The ingenious Arab who first stirred up the bean broth from coffee cherry seds never thought that his concoction will later stir the world. Launched from Arabia, through night long rituals to keep the worshippers awake, for teetotal muslims, it became an integral part of their religious and secular life. Coffee then became the national drink of Arabia. Thousands of Haj pilgrims making their annual trip to Mecca, goy converted to this new irresistible faith of coffee. The pilgrims often carried back coffee baens to their homes in India and

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